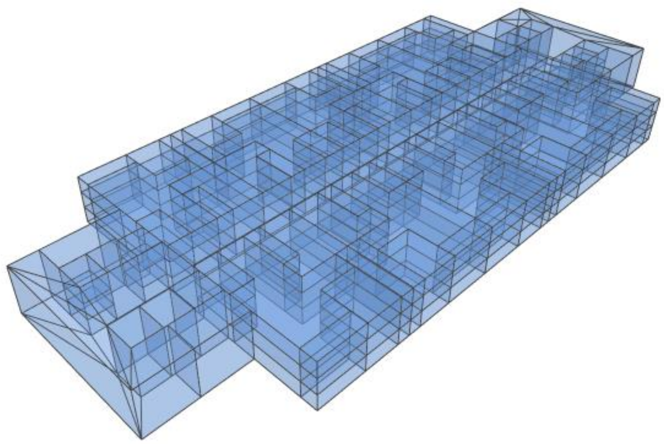
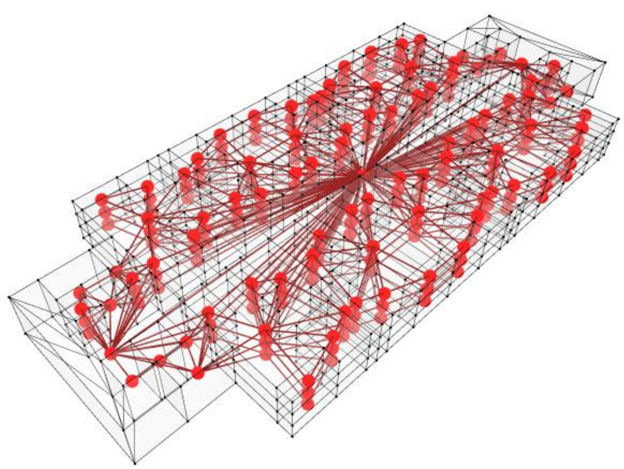
ASSIGNMENT 1- REPORT

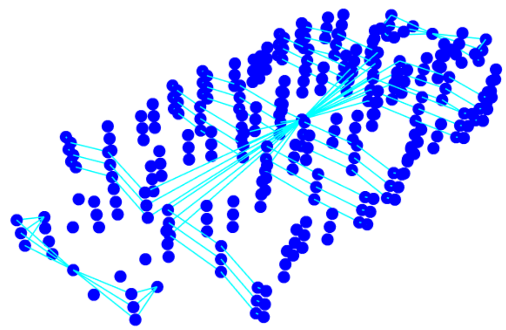
This assignment explored how architectural spaces can be translated into graph representations using Grasshopper and GraphML principles. The project focused on a residential building consisting of 10 apartment units on each floor. The building geometry was first developed in Grasshopper and then converted into a graph structure using nodes and edges to represent spatial relationships and circulation.



At the larger scale, each apartment was represented as a node, while the connections between apartments were represented through edges based on adjacency and accessibility. I then developed the model further by analyzing the rooms inside each apartment using the dual graph method, where rooms became nodes and shared boundaries or connections became edges. This allowed the building to be understood as a network of spatial relationships rather than only as geometry.



To create a more detailed circulation analysis, I introduced door apertures into the graph system. The apertures acted as transition points between spaces and generated a circulation graph showing possible movement paths throughout the building. The final visualization combined the apartment graph, room graph, and circulation graph together using different colors to distinguish between spatial categories and graph layers.



One of the main challenges during the project was controlling the complexity of the graph and accurately color coding the rooms. As the graph became denser, the visualization became difficult to read and the spatial classification was not always fully accurate. This made me realize that the graph still needs more semantic meaning and stronger control over how architectural elements are recognized. In future iterations, introducing more specific aperture categories, such as importing bathrooms, bedrooms, or circulation cores separately, could improve how the graph identifies and organizes spaces. Another challenge was computational performance, as the heavy geometry and large amount of graph data slowed down the workflow in TopologicPy. Future improvements would focus on optimizing the dataset and simplifying the graph structure while maintaining meaningful spatial relationships.